

Graph Theory - Unit 1: 100 Questions with Detailed Answers

Set 1: Graphs and Subgraphs, Isomorphism, Graph Matrices

1. Define a graph. Give an example.

A graph $G = (V, E)$ consists of a non-empty set V of vertices and a set E of edges.

2. What is a simple graph?

A simple graph is an undirected graph with no loops and no multiple edges.

3. Differentiate between a graph and a multigraph.

A graph has at most one edge between any pair of vertices, whereas a multigraph allows multiple edges.

4. What is a subgraph?

A subgraph of a graph $G = (V, E)$ is a graph $H = (V', E')$ where $V' \subseteq V$ and $E' \subseteq E$.